

Allison E. Grimsted
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EDUCATION

University of Michigan, College of Engineering, Ann Arbor, MI **August 2024 – Present**
PhD, Industrial and Operations Engineering (IOE), Operations Research concentration
Advanced to candidacy: April 2026
Advisor: Professor Brian Denton

Cornell University, College of Engineering, Ithaca, NY **August 2017 – May 2021**
Bachelor of Science, Operations Research and Engineering (ORIE), *Magna Cum Laude*
Minors: Mathematics; Biometry and Statistics
Research Advisor: Professor Shane Henderson

PROFESSIONAL EXPERIENCE

Johns Hopkins University Applied Physics Laboratory (JHU/APL) **Laurel, MD**
Associate Operations Research Analyst I, Decision Systems Group August 2021 – July 2024

- Developed optimization algorithms, large-scale simulations, and data analytics pipelines to support complex operational decisions for federal agencies and internal research & development.
- Led development and deployment of software for a federal agency, overseeing quality assurance and mentoring an intern.
- Frequently briefed government program manager to demonstrate prototypes and display results.
- Supported the federal COVID-19 response by building automated pipelines from numerous data streams into quality-controlled daily reports for government officials and public dissemination.

AWARDS

IOE Bonder Fellowship, University of Michigan 2025

- Awarded competitive department fellowship, which provides full support for two semesters.

Coin Award, JHU/APL 2022, 2023, 2024

- Awarded three times for leadership in research and development as an early career scientist.
- 5% of JHU/APL technical staff win a coin award annually.

Building Leaders, Accelerating Science and Technology Grant, JHU/APL 2022

- Awarded \$10,000 to fund portion of "Hybrid Bistatic Radar for In-Situ Debris Sensing."
- Less than 20% of grant applications awarded.

Nominee for Excellence in Quantitative Sciences, Centers for Disease Control 2022

- Nominated for this award based on models and data analytics developed for the federal COVID-19 response in education.

Omega Rho, Cornell University 2021

- Awarded to undergraduates in the top 15% of the ORIE major.

Engineering Learning Initiatives Undergraduate Research Grant, Cornell University 2020

- Awarded \$4,500 to fund summer research, "Dispatch Policies for Urgent/Emergent Calls in Air and Land Ambulance Operations."

Tau Beta Pi, Cornell University 2020

- Awarded to undergraduates in the top 12% of the engineering school.

WORKING PAPERS

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1. **Grimsted, A.**, Orfanoudaki, A., Denton, B. (2026) *Sample-Path Clustering for Context-Sensitive Markov Decision Processes*.

TECHNICAL REPORTS

1. JHU/APL COVID-19 Operations Team (weekly 2021-2023). *COVID-19 State Profile Reports*, U.S. Department of Health and Human Services. <https://healthdata.gov/browse?tags=covid-19-spr>.
2. JHU/APL COVID-19 Operations Team (daily/biweekly 2021-2023). *COVID-19 Community Profile Report*, U.S. Department of Health and Human Services. <https://healthdata.gov/Health/COVID-19-Community-Profile-Report/gqxm-d9w9>.
3. McGarity, M., Swan, B., Calloway, R., Baker, J., McShea, M., Beatty, P., **Grimsted, A.**, Hill, D., Howe, D., Jones, D., Matthew, K., Camacho, E., Brown, G., Llera Ortiz, J., Mekonnen, Y., Helmer, D. (2023). *Aeromedical Evacuation Human Performance Medical Mission Model Under Agile Combat Employment Environment – Summary Final Report*, JHU/APL Technical Communications, AOS-23-1509.
4. McGarity, M., Swan, B., Calloway, R., Baker, J., McShea, M., Beatty, P., **Grimsted, A.**, Hill, D., Howe, D., Jones, D., Matthew, K., Camacho, E., Brown, G., Llera Ortiz, J., Mekonnen, Y., Helmer, D. (2023). *Aeromedical Evacuation Human Performance Medical Mission Model Under Agile Combat Employment Environment – Analysis and Capability Gaps Assessment Report*, JHU/APL Technical Communications, AOS-23-1468.
5. **Grimsted, A.**, Frey, C. (2022). *Graph Based Analytic Model for Resilience Analysis*, JHU/APL Technical Communications, AOS-22-1208.
6. **Grimsted, A.**, Henderson, S. (2020). *Dispatch Policies for Urgent/Emergent Calls in Air and Land Ambulance Operations*, Engineering Learning Initiatives Report, Cornell University.

INVITED CONFERENCE PRESENTATIONS

1. **Grimsted, A.**, Piri, H., Denton, B. Bartell, A. (November 2026). *A Dynamic and Resource-Aware Risk Model for Hospital Admission of Patients with Hypertensive Disorders of Pregnancy*, INFORMS Annual Meeting.
2. **Grimsted, A.**, Orfanoudaki, A., Denton, B. (2026). *Sample-Path Clustering Optimization for Context-Sensitive Markov Decision Processes*, Michigan Student Symposium for Interdisciplinary Statistical Sciences.
3. **Grimsted, A.**, Orfanoudaki, A., Denton, B. (2026). *Sample-Path Clustering for Context-Sensitive Markov Decision Processes*, INFORMS Optimization Society Conference.
4. **Grimsted, A.**, Orfanoudaki, A., Denton, B. (2025). *Sample-Path Clustering for Context-Sensitive Markov Decision Processes*, INFORMS Annual Meeting.

INVITED SEMINAR PRESENTATIONS

1. **Grimsted, A.**, Swan Hovey, B. (2024). *Modeling Human Performance during a Simulated Aeromedical Evacuation Mission*, Modeling and Simulation Seminar, JHU/APL.
2. **Grimsted, A.** (2023). *Graph Based Analytic Model for Resilience Analysis*, Supply Chain Seminar, JHU/APL.

OTHER PRESENTATIONS

1. Flaherty, M., **Grimsted, A.**, Hingorani, R., Eddowes, D. (2022). *Hybrid Bistatic Radar for In-Situ Debris Sensing*, Building Leaders, Accelerating Science and Technology Capstone, JHU/APL.
2. **Grimsted, A.**, Maszkiewicz, L., Nichols, M., Raible, E., Gallagher, M. (2022). *National COVID-19 Response at JHU/APL*, Women in Technology Showcase, JHU/APL.

3. **Grimsted, A.**, Henderson, S. (2020). *Dispatch Policies for Urgent/Emergent Calls in Air and Land Ambulance Operations*, Engineering Learning Initiatives Summer Presentation, Cornell University.

POSTERS

1. Beatty, P., **Grimsted, A.** (2023). *Modeling Human Performance during a Simulated Aeromedical Evacuation Mission*, Asymmetric Operations Conference, JHU/APL.
2. **Grimsted, A.** (2022). *Graph Based Analytic Model for Resilience Analysis*, Asymmetric Operations Conference, JHU/APL.

INTELLECTUAL PROPERTY DISCLOSURES

1. Shin, L., **Grimsted, A.**, Behrendt, A. (2024). *Logistics Aid for Sustainment and Expeditionary Readiness*, PO7830.
2. McGarity, M., Swan, B., Calloway, R., Baker, J., McShea, M., Beatty, P., **Grimsted, A.**, Hill, D., Howe, D., Jones, D., Matthew, K., Camacho, E., Brown, G., Llera Ortiz, J., Mekonnen, Y., Helmer, D. (2023). *Human Performance Medical Mission Model (HPMx3)*, 7435-6400.

TEACHING

Teaching Assistant, Artificial Intelligence training (University of Michigan) Summer 2026

- Created course materials for teaching (deep) reinforcement learning with coding modules.

Teaching Assistant, ORIE 4741, Learning with Big Messy Data (Cornell University) Fall 2020

- Held weekly office hours to explain course content.
- Wrote solutions and grading guidelines for exams and homework.
- Managed a team of graders and resolved grade disputes.

Debate Coach (Evanston Township High School) Fall 2020 – Spring 2021

- Designed congressional debate curriculum for novice and experienced high school students during biweekly practices. Developed students' research skills.
- Judged at nation-wide tournaments and provided individualized feedback on students' speeches.

Grader, ORIE 3500/5500, Engineering Probability and Statistics II (Cornell University) Fall 2019

- Graded exams and homework for two sections.

SERVICE

Women in Science and Engineering Ann Arbor, Michigan
Graduate Student Mentor August 2026 – Present

- Host professional development events and social activities for undergraduate and graduate students.

Engage Detroit Detroit, Michigan
Volunteer, Optimization Team June 2025 – July 2025

- Supported multi-day outreach workshops introducing optimization to Detroit high school students through real-world examples and interactive games.

IOE Graduate Application Mentoring Program Ann Arbor, Michigan
Mentor September 2024 – Present

- Provide application guidance and answer questions about the department graduate program for prospective students to the IOE PhD program.